

# VR-Sets-ZFA — v1.0.2 (revision note)

Changes over v1.0.1:  $ZF^-$  on `OSetZFA`, a corrected IV.4, and the cost-of-quotient relation to VR-Sets (Brouwer Edition)

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**Scope.** This is the revision delta for VR-Sets-ZFA v1.0.2; Parts I–VII of v1.0.1 are unchanged except as stated below. Three changes: (A) the relation to VR-Sets (Brouwer Edition) is made explicit, recasting VR-Sets-ZFA’s role as the *quotiented twin* that measures the cost of the quotient; (B) the strict inclusion of §IV.4 is now *earned* rather than asserted, because (C) the  $ZF^-$  axioms are now machine-verified on `OSetZFA` itself. All Lean objects named below are in module `VRCycle.SetsZFA`; the build is green and sorry-free.

## 1 (A) Relation to VR-Sets (Brouwer Edition): the cost of the quotient

VR-Sets-ZFA was motivated (§I.2) by the observation that the inductive `PSet` foundation of VR-Sets makes non-well-foundedness impossible in `OSet` (`AFA_Refuted`, `quineAtom_impossible`), so a *coinductive* universe `OSetZFA` was built to obtain AFA. That motivation is now **historical**. VR-Sets has since been rebuilt on the Brouwer path (VR-Sets, Brouwer Edition) on a different carrier — the pointed graph `OpSet`, with identity carried as a *witnessed bisimulation and no quotient*. There the Quine atom is a one-vertex self-loop and Anti-Foundation is a theorem with axiom profile `[]` (fully axiom-free, `OpSet.afa`): a member is the same graph re-pointed, so nesting — including self-membership — is native, and decoration is almost definitional.

VR-Sets-ZFA’s actual, current role is the **quotiented twin** of that construction, and the pair is itself a result: the *same* content (Anti-Foundation) costs `[]` in the relational form (`OpSet`) and the full ceiling `[propext, Classical.choice, Quot.sound]` in the quotiented form (`OSetZFA`). Forming the quotient Skolemises the bisimulation’s existential witnesses — VR’s transit asymmetry — and uniqueness must select representatives by `Classical.choice` and run `PFuncor.M.bisim` through the choice-pulling destructor `dest`. This is the sharpest quantitative illustration of the transit asymmetry in the cycle: not “the quotient pulls choice somewhere”, but one theorem, its profile before and after the quotient.

*Remark 1.1* (The two universes, read together). We read the two works together as the equivalence  $\text{OSetZFA} \simeq \text{OpSet}/\approx$ , with VR-Sets-ZFA the quotient of which the Brouwer Edition is the choice-free floor. **This equivalence is a prose-level reading, not yet machine-checked** (in the modality of Conjecture IV.2); its formalisation — the `M.corec` unfolding of a graph, descent through the quotient, and bijectivity on the  $\approx$ -classes — is a single `Bridge` file, a candidate for v1.1.

## 2 (C) ZF<sup>-</sup> axioms on OSetZFA

v1.0.1 proved AFA on OSetZFA and a strict  $\in$ -preserving embedding of the well-founded universe, but it did *not* verify the ZF axioms on OSetZFA itself — only  $\emptyset$ , singleton and extensionality were present. They are now all machine-checked, so that OSetZFA is a model of ZF minus Foundation (Foundation being supplied by AFA).

**Theorem 2.1** (Pairing, Union, Power).  $z \in \{a, b\} \leftrightarrow z = a \vee z = b$  (*mem\_pair*);  $z \in \bigcup x \leftrightarrow \exists y, y \in x \wedge z \in y$  (*mem\_sUnion*);  $z \in \wp x \leftrightarrow z \subseteq x$  (*mem\_powerset*). Each is built at the *CoPSet* level as a single *mk $\alpha$ A* touching only the top branching levels (the coinductive structure affects only equality, supplied by *isBisim\_Equiv* / *bisim\_imp\_Equiv*); the workhorse *CoPSet.mk\_equiv\_of\_coverage* reduces each well-definedness obligation to a child-coverage check. Power set is a verbatim *PSet.powerset* *calque* — no special difficulty from the coinductive quotient.

**Theorem 2.2** (Separation and Replacement — operational). For a predicate  $p : \text{OSetZFA} \rightarrow \text{Prop}$ ,  $z \in \{w \in x \mid pw\} \leftrightarrow z \in x \wedge pz$  (*mem\_sep*); for a function  $F : \text{OSetZFA} \rightarrow \text{OSetZFA}$ ,  $z \in \{Fy \mid y \in x\} \leftrightarrow \exists y, y \in x \wedge Fy = z$  (*mem\_image*). Because  $p$  and  $F$  are first-class objects over the quotient, they are automatically  $\approx$ -invariant: the collapse of the first-order Separation/Replacement schemas into single quantified-over-an-object statements is the operational reification of describability, not a generic type-theoretic accident.

**Theorem 2.3** (Infinity). With the von Neumann successor  $\text{succ } x := \text{insert } x \ x = x \cup \{x\}$  and  $\omega$  the set of von Neumann naturals:  $\emptyset \in \omega$  and  $\forall x, x \in \omega \rightarrow \text{succ } x \in \omega$  (*infinity*);  $z \in \omega \leftrightarrow \exists n, z = \llbracket \text{ofNat } n \rrbracket$  (*mem\_omega*). The naturals match VR’s  $O_n$  and *mathlib*’s *ZFSet.omega*.

*Remark 2.4* (OSetZFA models ZF<sup>-</sup>, at the M-type ceiling). With Extensionality,  $\emptyset$ , Pairing, Union, Power, Separation, Replacement and Infinity verified, and Foundation supplied as AFA, every member of OSetZFA obeys the ZF<sup>-</sup> axioms as machine-checked theorems. All witnesses sit at [*propext*, *Classical.choice*, *Quot.sound*] — the M-type ceiling inherited from *PFunctor.M.dest*, not the set theory (the choice-free counterparts live on the *OpSet* carrier, §1). **Honest boundary:** each axiom holds in OSetZFA as a theorem; OSetZFA is not itself given a first-order satisfaction relation — that “ $\models \text{ZF}^-$ ” reading stays metatheory, as elsewhere in the cycle.

## 3 (B) Correction to §IV.4 — ZFC $\subsetneq$ ZFA earned

v1.0.1’s §IV.4 stated “ZFC  $\subsetneq$  ZFA”. What it had machine-checked was the strict inclusion of *carriers with membership*: *embedOSet* is injective (*embedOSet\_injective*) and  $\in$ -preserving (*embedOSet\_mem*), and the Quine atom witnesses non-surjectivity (*quineAtom\_not\_in\_range\_embedOSet*). That is an inclusion of *universes*, not yet of *theories*: “ZFA universe” was not fully earned because the ZF<sup>-</sup> axioms were not verified on OSetZFA. With §2 above, the gap is closed.

**Theorem 3.1** (§IV.4, corrected). *OSetZFA* is a model of ZF<sup>-</sup> (§2); *OSet* is the grounded (well-founded) fragment, and  $(\text{OSet}, \in) \subsetneq (\text{OSetZFA}, \in)$  strictly, with the Quine atom the explicit witness of the strictness. Hence the inclusion is of universes both modelling ZF<sup>-</sup>, the well-founded one properly inside the one admitting cycles — the earned form of “ZFC  $\subsetneq$  ZFA”.

The boundary of the previous remark applies: this is an inclusion of machine-checked models, not a metatheoretic theorem-set inclusion (which stays, as throughout the cycle, in the metatheory).

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## References (new)

- V. Reznik, *VR-Sets: An Operational Set Theory Unifying ZFC and ZFA (Brouwer Edition)*. Zenodo, 2026. DOI 10.5281/zenodo.20643407, v1.1.0 (concept 10.5281/zenodo.20303535).
- The VR Cycle Lean 4 formalisation, <https://github.com/inventor1975/VRCycle>, modules `VRCycle.SetsZFA` (this work) and `VRCycle.SetsOp` (Brouwer Edition).